No Patient Left Behind:
Bridging Language Barriers in Rhode Island's EHR Systems

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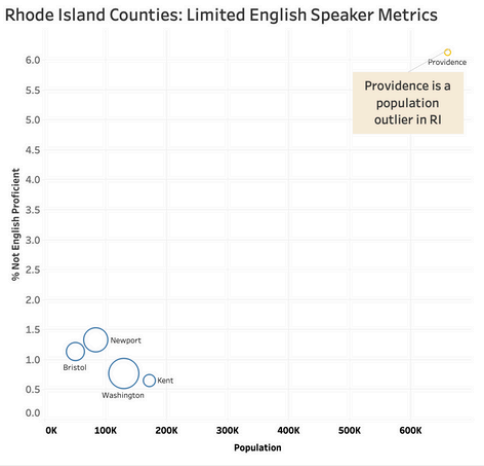
Background:

- Many patient portals in RI present complex medical jargon only in English
- Providence County has 660K residents, 17% Latino, 6.1% limited English proficient (LEP)



- SQL and jargon dictionary extract EHR
- ↓ Calls about test results
- ↑ Portal logins & feature use
- ↑ Patient comprehension
- ↓ Provider burden

Evaluation & Impact

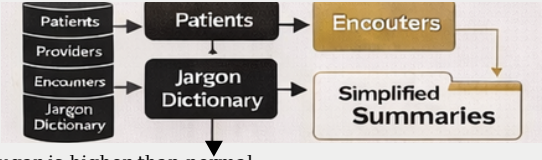
- At least 80% of patients demonstrate improved task-based comprehension after translation
- At least 20% increase in portal usability and satisfaction
- Implement a 4-6 week pilot prior to post pilot evaluation

Objective

- Develop an equity-driven patient portal that translates raw medical jargon into simplified, bilingual outputs.
- Pilot in Providence County to bridge language barriers for LEP Spanish and Portuguese speakers

FHIR-compatible translation layer

Translational Workflow



- Your blood sugar is higher than normal
- Take metformin twice a day with food
- Avoid soda, cake, and candy

- O seu açúcar no sangue está mais elevado do que o desejado
- Tome metformina duas vezes por dia, às refeições.
- Evite refrigerantes, bolos e doces.

- SQL and jargon dictionary extraction from EHR
- Translation and simplification logic for lab results
- Operationalizing health equity improves access by reducing disparities, lowering costs, and building patient trust